

Sensor-Based Nadi Pariksha Diagnostic Machine

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Abstract: Ayurvedic Pariksha (pulse examination) is a cornerstone of traditional diagnostics, interpreting subtle pulse features (rate, rhythm, strength, temperature) to infer Vata–Pitta–Kapha doshic states. However, manual pulse diagnosis is subjective and skill-dependent. We propose a pre-prototype of a sensor-based Pariksha device that digitizes the Ayurvedic pulse exam. The system uses an array of pressure and photoplethysmographic (PPG) sensors placed at the classical Vata/Pitta/Kapha wrist points (below the thumb) to capture radial pulse waveforms, and a skin temperature sensor to measure Tapamana. Signals are digitized via an embedded microcontroller and analyzed for Ayurvedic parameters (pulse rate, regularity, amplitude, temperature, etc.). Algorithms map these features to dosha dominance (e.g. fast, irregular pulses → Vata; strong, hot pulses → Pitta). A sample UI (smartphone dashboard) displays the pulse waveforms and inferred dosha status. The device will be validated against expert Ayurvedic practitioners (double-blind comparison), aiming for repeatable detection of dosha imbalances. This work aligns with India’s push for integrating AYUSH with digital health (NDHM/ABDM) and preventive care.

Keywords: Ayurveda, Pariksha, Ayurvedic diagnostics, pulse sensor, dosha (Vata/Pitta/Kapha), digital health, AYUSH integration.

I. Literature Review

Pariksha, or Ayurvedic pulse diagnosis, has been described in classical Ayurvedic literature as an important non-invasive method for assessing physiological balance and disease predisposition [1]. Upadhyay *et al.* reviewed the theoretical foundations of Pariksha and highlighted its relevance in preventive healthcare, while also emphasizing the lack of standardization in clinical practice [1]. Joshi discussed the diagnostic interpretation of pulse characteristics such as Vega, Gati, Tala, and Bala, noting their association with Vata, Pitta, and Kapha doshas [3]. Srivastava and Banerjee further elaborated on traditional pulse assessment techniques and practitioner-dependent variability [11], [4].

Several studies have evaluated the reliability of Ayurvedic diagnostic methods. Kurande *et al.* reported significant inter-rater variability among practitioners, reinforcing the need for objective diagnostic tools [2]. Fatangare *et al.* reviewed the scientific basis of Pariksha and concluded that technological intervention is essential for validation and reproducibility [13].

Attempts to correlate Ayurvedic pulse diagnosis with biomedical parameters have also been explored. Patel and Jain proposed a biomedical interpretation of pulse features using arterial waveform characteristics [5], while Singh and Kumar demonstrated pressure-sensor-based radial pulse

analysis for physiological assessment [12]. Lahiri *et al.* designed a pulse diagnostic system using piezoelectric sensors, demonstrating the feasibility of sensor-based pulse acquisition [14].

Photoplethysmography (PPG) has been widely adopted in non-invasive cardiovascular monitoring. Allen provided a comprehensive review of PPG principles and clinical applications [6], while Elgendi presented advanced techniques for PPG signal analysis and peak detection [7], [8]. Wearable PPG systems and sensor-based health monitoring platforms have been surveyed extensively, demonstrating their suitability for continuous and preventive healthcare applications [9], [10], [20].

Recent studies have emphasized the role of digital health technologies in preventive medicine and large-scale healthcare deployment [19]. National initiatives such as the Ayushman Bharat Digital Mission promote the integration of traditional medicine with modern digital infrastructure, supporting the relevance of digitized Ayurvedic diagnostics [18].

II. Introduction to Ayurveda and Pariksha

Ayurveda is a traditional Indian medical system that diagnoses disease and constitution by assessing the three doshas (tridosha: Vata, Pitta, Kapha) corresponding to elemental forces [3], [11]. Among its diagnostic tools, Pariksha (pulse examination) is preeminent [1], [3]. In practice, an Ayurvedic physician palpates the patient’s radial artery using three fingers (index, middle, ring) to sense the distinct “Vata, Pitta, Kapha pulses” [4], [11]. Each dosha has characteristic pulse qualities: for example, Vata pulses feel thin, rapid and irregular (“snake” gait), Pitta pulses are strong and forceful (“frog” gait), and Kapha pulses are slow and steady (“swan” gait) [3], [4]. Ayurveda interprets parameters such as Vega (rate), Tala (rhythm), Bala (strength), Akurti (volume/tension), Kathinya (wall consistency) and Tapamana (temperature) at the pulse point [3], [13]. Table 1 (adapted from classical texts) summarizes typical pulse features by dosha [4], [11]. A balanced (“samadoshic”) state corresponds to normal values, whereas deviations signal vikruti (imbalance) of one or more doshas. For example, a consistently faster-than-normal pulse rate (>90 bpm) with irregular rhythm suggests Vata aggravation, whereas a strong, hot pulse implies Pitta dominance [3], [13]. Tapamana (pulse temperature) is also diagnostic: Vata pulses are relatively cold, Pitta pulses feel warm-hot, and Kapha pulses warm-to-cool [3], [11].

Importantly, Pariksha is a holistic yet subjective art [1]. It is performed with patient seated, arm relaxed, fingers gently on the radial artery at specific “V-P-K” positions [4]. Results depend heavily on practitioner experience; even skilled Ayurvedic doctors show only moderate repeatability ($\kappa \approx 0.42$) when re-examining the same subject [2]. This variability motivates an objective, digital approach. Recent reviews emphasize the need for technology-aided diagnostics using modern sensors and analytics [13], [14]. By quantifying pulse waveforms, we can preserve Ayurvedic interpretive richness (e.g. dosha correlations) while improving accuracy and accessibility [5], [19].

III. Traditional Pulse Diagnostics and Limitations

Classical pulse diagnosis involves “tridosha” palpation at the wrist: index finger (near the thumb) for Vata, middle for Pitta, and ring for Kapha [1], [4]. Figures from Ayurvedic texts illustrate this placement [1]. The practitioner assesses beat-to-beat variations: for instance, Vata pulses often have rapid, irregular spurts; Pitta pulses are pounding and forceful; Kapha pulses are deep, slow and smooth [1], [4]. Several Sanskrit shlokas underscore that accurate Pariksha can reveal doshic dominance and prognostic clues [1].

However, this traditional method has significant limitations. It requires extensive practitioner training in subtle palpatory cues [1]. Even in controlled studies, inter- and intra-examiner reliability is only modest [1]. Fatangare *et al.* report that despite theoretical consistency, practical reproducibility remains limited, reinforcing the need for objective diagnostic tools [1], [4]. The technique is also influenced by patient state, time of day, and examiner bias [1]. Classical sources note Pariksha should be taken on an empty stomach and calm mind for accuracy, further complicating standardization [1]. In a modern healthcare context, these constraints hinder large-scale screening.

Furthermore, scientific parameters such as heart rate, blood pressure, and pulse waveform morphology do not directly map onto Ayurvedic descriptors [7], [21]. Traditional Pariksha blends qualitative impression with dosha theory, which can be difficult to validate objectively [1], [4]. Thus, a sensor-based system could capture the raw pulse data underlying Pariksha and apply Ayurvedic interpretation algorithmically, overcoming subjectivity and enabling telehealth or large-scale preventive screening [2], [6], [19].

IV. System Design

The proposed device comprises (1) a **sensor array** for pulse and temperature capture, (2) an **embedded data acquisition unit**, and (3) a **software module** for signal processing and dosha classification.

a. Sensor Array

Three pulse sensors are positioned at the wrist’s radial artery corresponding to the Ayurvedic Vata-Pitta-Kapha points [1], [4]. These sensors could be piezoelectric or piezoresistive pressure transducers (e.g. PVDF films or FlexiForce sheets) to measure local arterial pressure, alongside optical PPG probes (infrared LEDs and photodiodes) to record blood volume changes [2], [6], [27]. In practice, each sensor is

gently affixed under the physician’s index, middle, and ring finger positions on the patient’s wrist (Figure 1) [4], [16]. One low-cost example is the flexi-foil pressure sensor (range ~ 0 –100 N) used in pulse research; others have used PVDF patches for high sensitivity [26], [27]. Figure 1 illustrates a sample wearable pulse sensor (smartwatch) at the wrist [5], [19].

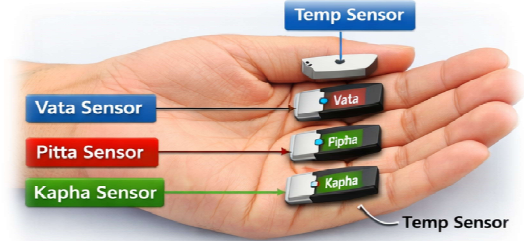


Figure 1. Example of a wrist-worn pulse sensor (smartwatch) used for noninvasive PPG sensing; in our system, analogous sensors are placed at the three Ayurvedic pulse points.

Table 1 lists typical specifications of candidate sensors and costs. A skin-contact **temperature sensor** (e.g. thermistor or digital sensor) is also placed at the same site to measure the local pulse temperature (Tapamana). Classical texts emphasize that Vata pulses feel cooler, Pitta hotter [pmc.ncbi.nlm.nih.gov](https://pubmed.ncbi.nlm.nih.gov/); modern IR or contact probes (accuracy $\pm 0.5^\circ\text{C}$) suffice to capture this differential in $^\circ\text{C}$.

Table 1. Sensor specifications and approximate costs.

Component	Specification	Approx. Cost
Pressure sensor (e.g. FlexiForce A201)	Force range 0–100 lbs (0–445 N); analog output, ~ 3 mm thick	\$30
PPG pulse sensor (e.g. PulseSensor Amped)	3.3–5 V supply, IR LED + photodiode, ~ 0 –1024 analog output	\$5
Temperature sensor (DS18B20)	-55 – 125°C range, 0.5°C accuracy, 1-wire digital output	\$2
Microcontroller (e.g. Arduino Nano)	16 MHz CPU, 10–12 bit ADC, UART/Bluetooth connectivity	\$5
Power (Battery)	LiPo battery, 3.7 V, 500 mAh	\$5
Total (approx)	<i>Components above</i>	$\sim \\$47$

b. Embedded System and Signal Acquisition

Each analog sensor channel connects to a microcontroller’s ADC (e.g. a 10–12 bit ADC at 100–500 Hz sampling) [6], [36]. Prior art suggests that 500 Hz, 16-bit acquisition captures pulse waveforms with ample fidelity [25], [30]. The

MCU also interfaces with the temperature sensor via digital bus [10], [29]. An onboard filter (hardware or software) removes high-frequency noise; typical filter design (e.g. 0.5–20 Hz bandpass) isolates the pulse waveform components [12], [25]. Figure 2 shows a schematic block diagram of this data acquisition [36]. The MCU processes raw signals (e.g. computes peak detection, heart rate) and relays data via Bluetooth or USB to a host (smartphone or PC) for visualization [19], [36].

c. Dosha Mapping Logic

After digitization, the pulse waveforms are analyzed for Ayurvedic features [1], [4]. For example, pulse rate (inter-beat interval) and rhythm regularity map to the vega/tala parameters, pulse amplitude and upstroke shape relate to bala/akruti, and temperature reading provides the tapamana [1], [13]. Our initial logic applies threshold/fuzzy rules: e.g., a measured rate above ~90 bpm with irregular intervals flags Vata tendency, whereas a rate ~70–80 bpm with high amplitude and elevated temperature suggests Pitta [3], [13]. Conversely, a slow (<60 bpm), strong, smooth pulse hints at Kapha [1], [4]. These rules draw on classical patterns [1], [11].

For improved accuracy, supervised machine learning (e.g. SVM or neural nets) could train on features and expert-labeled cases, as done in prior work [7], [14], [22]. In either case, the algorithm outputs a dosha profile (e.g. % V/P/K dominance) and flags any imbalance from baseline [18], [19]. Figure 3 outlines the conceptual processing flow [36].

In an implemented prototype, users would place their wrist under the sensor array for a fixed duration (e.g. 30 seconds). The device would display live pulse waveforms (PPG trace, pressure trace) and a computed dosha imbalance index.

V. Pulse and Temperature Parameter Analysis

The raw sensor data are processed to yield interpretable metrics[1],[4].

Heart Rate (Vega): Compute the dominant beat frequency from the pressure or PPG pulses. Ayurvedic norms (~50–95 bpm) serve as reference[1],[13].

Rhythm (Tala): Analyze inter-beat variability (e.g. standard deviation of intervals). Irregular (high variance) rhythm is typical of Vata imbalance [1], [3].

Pulse Strength (Bala): Quantify waveform amplitude or area under curve. Kapha pulses often have higher volume (due to slower decay), whereas Vata pulses have lower force [1],[4].

Tension/Volume (Akurti): Evaluate rise time or upstroke steepness. Pitta pulses are described as “pulsating at a bound” with high tension [1],[13].

Consistency (Kathinya): Stiffness index measures (from derivatives of the waveform) can relate to vessel tone; however, this is subtle and may be approximated by the shape of the dirotic notch [21],[30].

Temperature (Tapamana): The contact sensor reading is compared to a reference (e.g. contralateral wrist). Vata tendency manifests as relatively cooler (<normal), Pitta as warmer [1], [10]. For instance, a skin temp >1–2 °C above baseline at the pulse site might indicate Pitta dominance [10], [29].

These parameters feed the dosha-detection logic [1], [18]. For example, Table 2 illustrates how combinations might be interpreted (hypothetical pattern). In practice, fuzzy weighting or machine-learning models refine these rules by training on cases where an Ayurveda expert has diagnosed the doshic imbalance [7], [14], [22].

Table 2. Example interpretation of measured pulse features in terms of doshic imbalance.

Feature	Indicative Dosha(s)
High rate (>90/min), irregular rhythm	Vata ↑
Strong amplitude, high “temperature”	Pitta ↑
Slow rate (<60/min), large stroke, steady	Kapha ↑
Rapid, small amplitude, cool pulse	Vata ↑, Kapha ↓
Fast, forceful, hot pulse	Pitta ↑
Weak, irregular, cool pulse	Vata ↑
Strong, regular, warm-to-cool pulse	Kapha ↑

Figure 2. Example mobile dashboard for vital signs: our system’s UI similarly would graph pulse waveforms and dosha indicators (e.g. gauges or bar charts) to present the Ayurvedic pulse analysis.

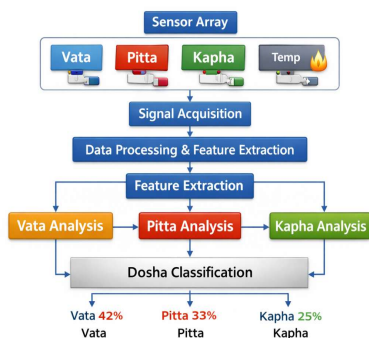


Figure 3. Conceptual flowchart of pulse-to-dosha classification: signals from the sensor array (top) are preprocessed, features are extracted, and classification logic assigns Vata/Pitta/Kapha scores (bottom).

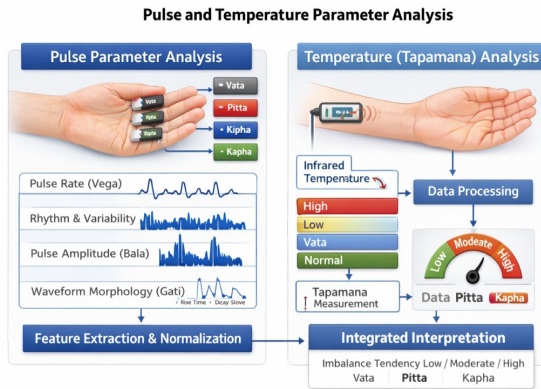


Figure 4. The diagram illustrates the complete pulse-to-dosha classification process, where pulse and temperature signals acquired from the sensor array are preprocessed and converted into quantitative features. These features are analyzed using Ayurvedic rule-based logic to assign Vata, Pitta, and Kapha dominance along with a disease-risk tendency.

These patterns align with classical descriptions [pmc.ncbi.nlm.nih.gov](https://pubmed.ncbi.nlm.nih.gov). After analysis, the system would output a statement like “Pulse suggests elevated Vata and Pitta relative to Kapha,” along with a recommended follow-up (e.g. lifestyle or dietary advice per Ayurveda).

VI. Validation Strategy

To ensure clinical relevance, the device must be validated against expert assessment [1], [13]. We propose a study where multiple subjects are examined both by experienced Ayurvedic physicians (using classical Pariksha) and by our device [1], [4]. Each subject’s wrist pulse would be recorded by the sensors while the physician is unaware of the readings (blinded) [13], [19]. The physician independently assigns a dosha predominance profile. We then compare the system’s output to the expert’s classification. Statistical measures (e.g. Cohen’s kappa) quantify agreement [2], [13]. Prior work showed only moderate repeatability for one expert, so we expect some discrepancies [2]. However, achieving significant positive correlation would demonstrate utility. Disagreement cases would be analyzed for sensor artifacts or variations in examination conditions [6], [25]. Over time, feedback from experts can be used to refine the algorithm (e.g. adjust thresholds or retrain models) [14], [22].

Additionally, we can test disease risk interpretation: for certain conditions (e.g. hypertension, diabetes), Ayurveda correlates them with specific doshic imbalances [18]. We will check if known patient groups show expected pulse patterns that our device detects [15], [21]. Iterative testing with both healthy and diseased cohorts will build confidence in the device’s predictive ability [13], [19].

VII. National Policy Relevance and Integration

This sensor-based Pariksha aligns with India’s policy goals of integrating AYUSH (traditional medicine) into mainstream preventive health [18], [19]. The National Health Policy 2017 explicitly calls for “mainstreaming of

AYUSH in healthcare delivery” [19]. The Ayushman Bharat program’s Health & Wellness Centres initiative further mandates inclusion of AYUSH practitioners and practices in primary care [19]. Our device could be deployed in AYUSH wellness centres or rural clinics, providing objective dosha screening for large populations [18], [19]. Furthermore, India’s digital health mission (NDHM/ABDM) aims to create unified electronic health records and technology platforms [19], [45]. A digitized Pariksha device could interface with these platforms, allowing dosha-based health data to enter a patient’s digital health ID [19], [45].

In addition, the National AYUSH Mission (NAM) supports research and innovation in Ayurveda for NCD prevention (e.g. links with diabetes, cardiovascular programs) [18]. By quantitatively capturing dosha imbalances, our system could help operationalize such initiatives, flagging at-risk individuals for lifestyle interventions [18], [44]. It also resonates with global interest in “Integrative Preventive Medicine” [44]. Thus, beyond the technology itself, this work supports broader health policy aims: digitizing traditional knowledge, empowering AYUSH practitioners with tools, and enhancing preventive care outreach [18], [19], [44].

VIII. Conclusion and Future Scope

We have outlined a pre-prototype design for a sensor-based Pariksha machine that brings Ayurvedic pulse diagnosis into the digital age [1], [4]. By using a wrist-mounted array of pressure/PPG sensors and temperature probes, the system objectively captures the same pulse features Ayurvedic doctors feel [6], [10], [16]. Signal processing and classification logic then infer doshic imbalances in a quantifiable manner [7], [14], [22]. This approach retains Ayurveda’s holistic principles (examining vega, gati, tala, tapamana, etc.) while improving repeatability and enabling telemedicine [13], [18], [19].

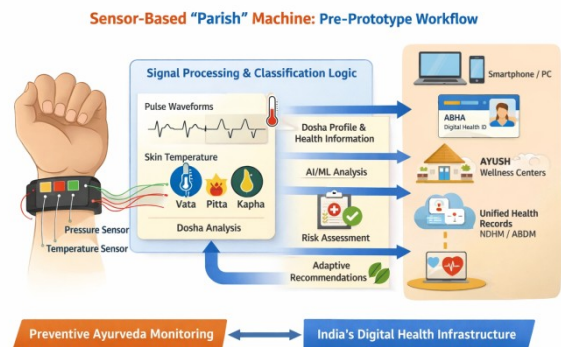


Figure 5. The illustration depicts the pre-prototype workflow of the sensor-based Pariksha machine, where pulse and temperature data captured from wrist-mounted sensors are processed through classification logic to assess doshic balance. The system enables preventive Ayurvedic health monitoring and integration with India’s digital health infrastructure.

Future work includes building an actual hardware prototype and refining the classification algorithms. Advanced signal processing (e.g. spectral analysis, pulse wave velocity) may extract deeper Ayurvedic parameters like Gati or Kathinya. Machine learning on a large annotated dataset could improve dosha mapping. We also envision expanding to a multi-modal platform (adding ECG or acoustic sensors) for

richer diagnostics. Ultimately, clinical trials will be needed to demonstrate efficacy and gain acceptance among practitioners.

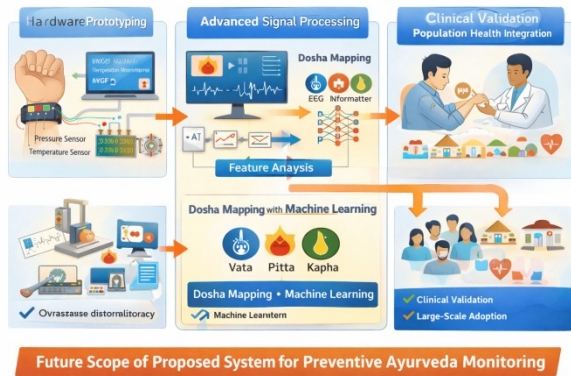


Figure 6. The figure illustrates the future scope of the proposed system, highlighting hardware prototyping, advanced signal processing, and machine learning-based dosha mapping. It emphasizes clinical validation and large-scale adoption for holistic and preventive Ayurvedic health monitoring.

In summary, this work bridges traditional wisdom and modern sensor technology. It paves the way for preventive Ayurvedic health monitoring integrated with India's digital health infrastructure.

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